

GREEN'S THEOREM

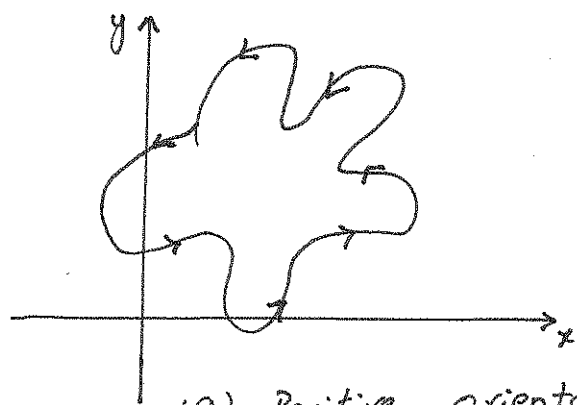
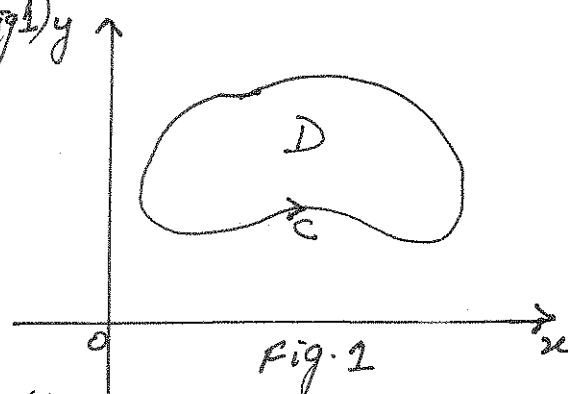
INTRODUCTION: Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C . (see fig 1)

In stating Green's Theorem

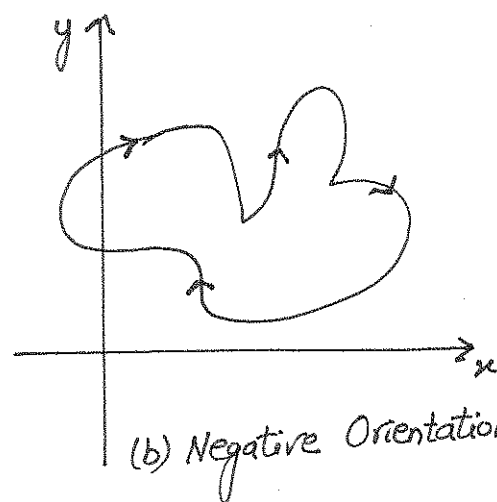
we use the Convention that

the Positive orientation of a simple closed curve ' C ' refers

to a single counterclockwise traversal of C . Thus if ' C ' is given by the vector function $\vec{r}(t)$, $a \leq t \leq b$, then the region D is always on the left as the point $\vec{r}(t)$ traverses C .



(a) Positive orientation



(b) Negative Orientation

STATEMENT OF GREEN'S THEOREM

Let ' C ' be a positively oriented, Piecewise-smooth, Simple closed curve in the plane and let D be the region bounded by ' C '. If ' P ' and ' Q ' have continuous partial derivatives on an open region that contains D , then

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

REMARK The notation

$$\oint_C P dx + Q dy \text{ or } \oint P dx + Q dy$$

is sometimes used to indicate that the line integral is calculated using the positive orientation of the closed curve C .

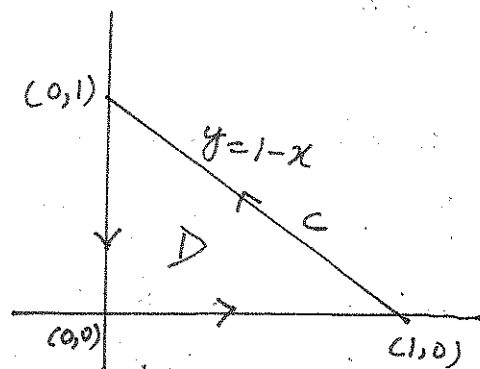
Another notation for the positively oriented boundary curve of D is ∂D , so the equation in Green's Theorem can be written as

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{\partial D} P dx + Q dy$$

EXAMPLE Evaluate $\int_C x^4 dx + xy dy$, where ' C ' is the triangular curve consisting of the line segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$ and from $(0,1)$ to $(0,0)$

Solution:. The region D enclosed by ' C ' is simple and ' C ' has positive orientation. If we let

$P(x,y) = x^4$, $Q(x,y) = xy$ then we have



$$\begin{aligned}
 \int_C x^2 dx + xy dy &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_0^1 \int_0^{1-x} (y - 0) dy dx \\
 &= \int_0^1 \left[\frac{1}{2} y^2 \right]_0^{1-x} dx = \frac{1}{2} \int_0^1 (1-x)^2 dx \\
 &= -\frac{1}{6} (1-x)^3 \Big|_0^1 = \frac{1}{6}
 \end{aligned}$$

EXAMPLE Evaluate $\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy$
 where 'C' is the circle $x^2 + y^2 = 9$ oriented counter clockwise.

Solution:. The region D is bounded by 'C' is the circle $x^2 + y^2 = 9$. So let's change to polar coordinates after applying Green's Theorem

$$P = 3y - e^{\sin x}, \quad Q = 7x + \sqrt{y^4 + 1}$$

$$\frac{\partial P}{\partial y} = 3, \quad \frac{\partial Q}{\partial x} = 7$$

$$\begin{aligned}
 \oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy &= \iint_D \left[\frac{\partial}{\partial x} (7x + \sqrt{y^4 + 1}) - \frac{\partial}{\partial y} (3y - e^{\sin x}) \right] dA \\
 &= \iint_D (7 - 3) r dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 4 r dr d\theta = 4 \int_0^{2\pi} d\theta \int_0^3 r dr = 36\pi
 \end{aligned}$$

